Lab13

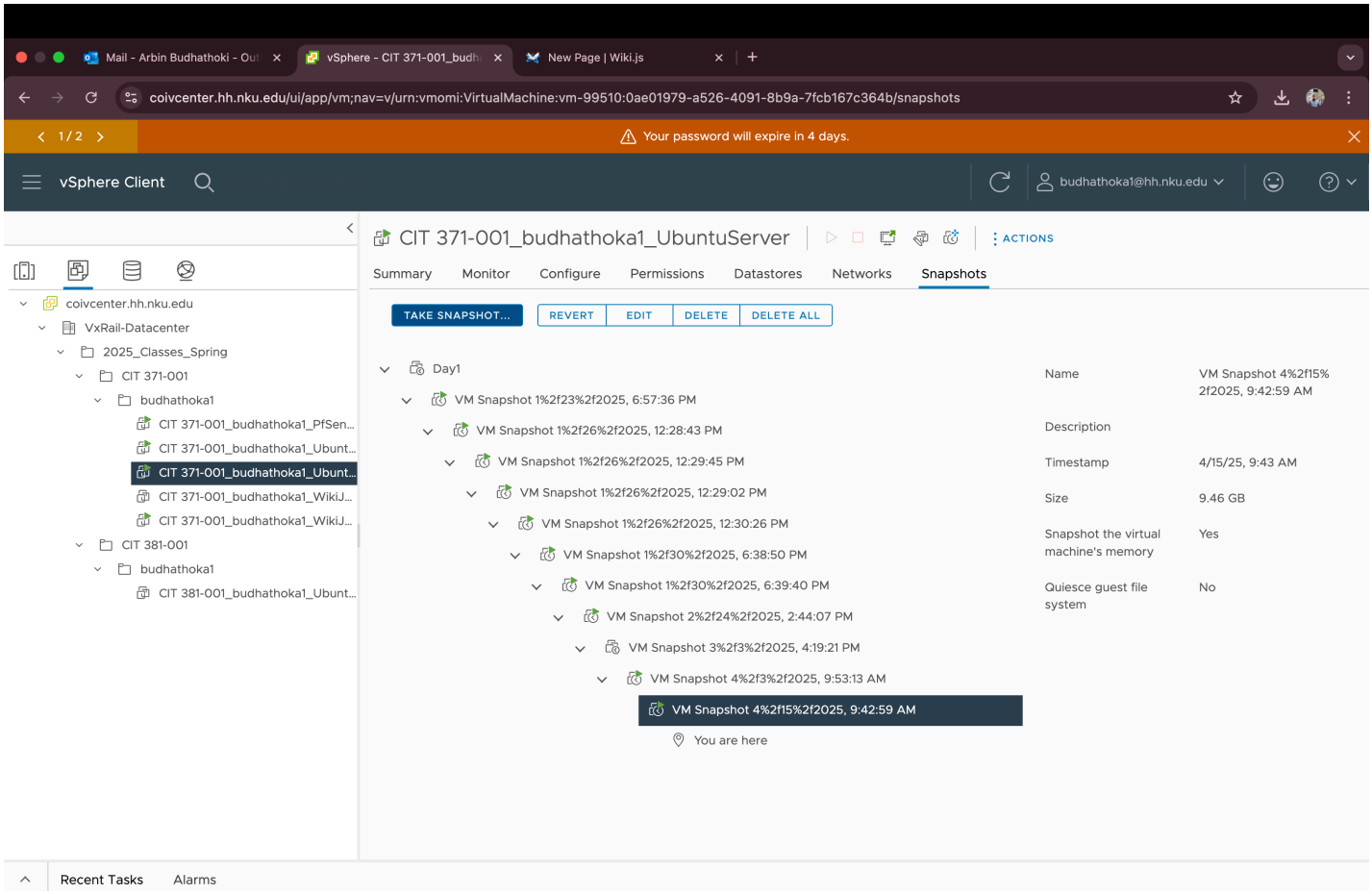
Lab 13: Backups and Cron Jobs

Overview:

In this lab, we explored two essential aspects of Linux system administration: tar and crontab. The tar utility is used for creating file archives, often for backup purposes. It allows combining multiple files into a single file (a tarball) that can later be compressed or extracted. It's a widely used tool for maintaining backups and transferring collections of files in a structured format. crontab is a utility for scheduling automated tasks in Unix-based systems. It lets users define scripts or commands to run at specific times or intervals. This is particularly useful for regular system maintenance tasks such as updates, cleanups, or monitoring reports without manual intervention.

Procedure:

Part0: Snapshot



Part2: Backups

→ sudo mkdir /backups

→ touch /home/student/backup1.txt /home/student/backup2.txt /home/student/backup3.txt

→ tar -cf /backups/stu_backup.tar /home/student/backup1.txt /home/student/backup2.txt /home/student/backup3.txt

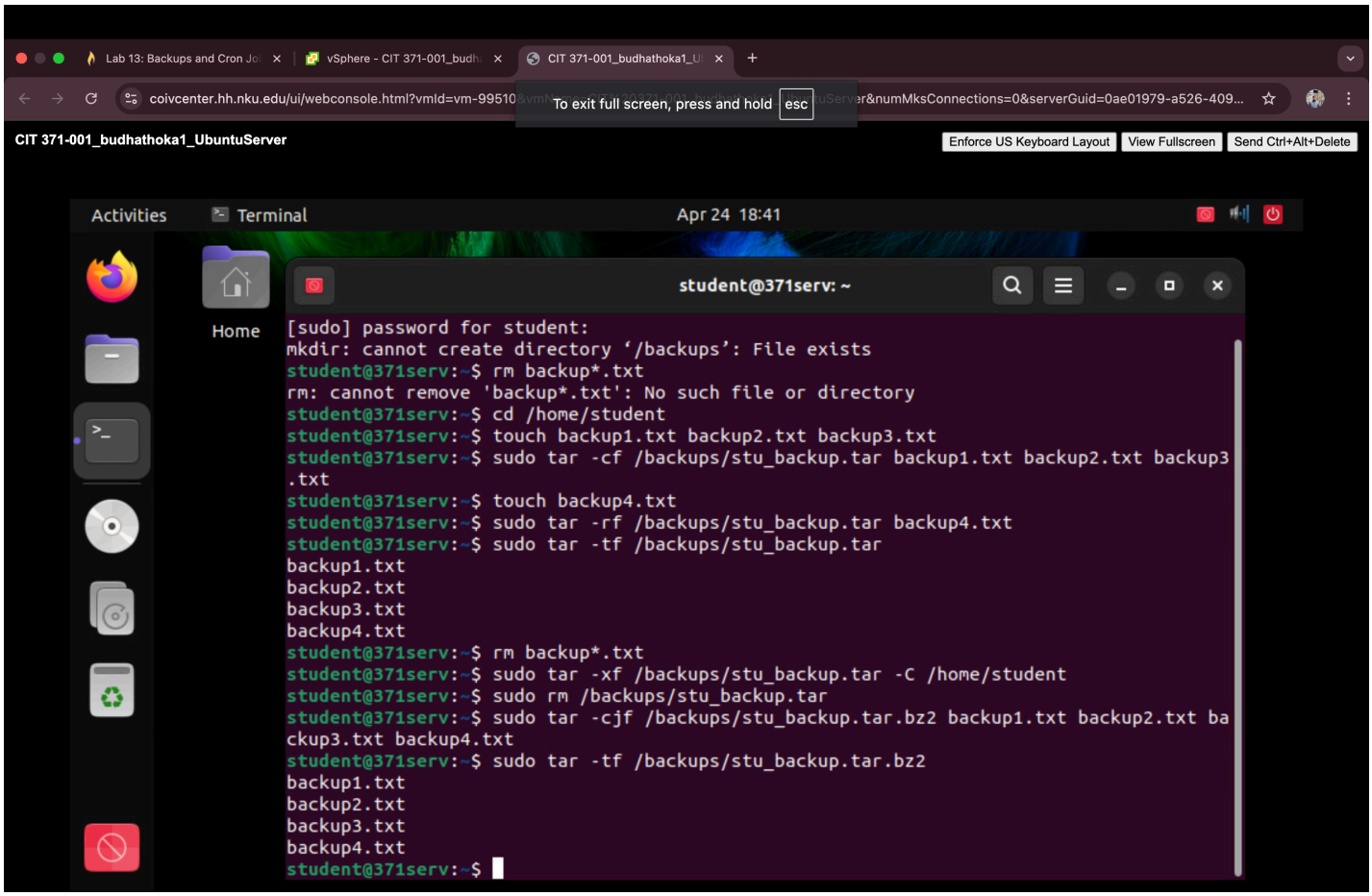
→ touch /home/student/backup4.txt

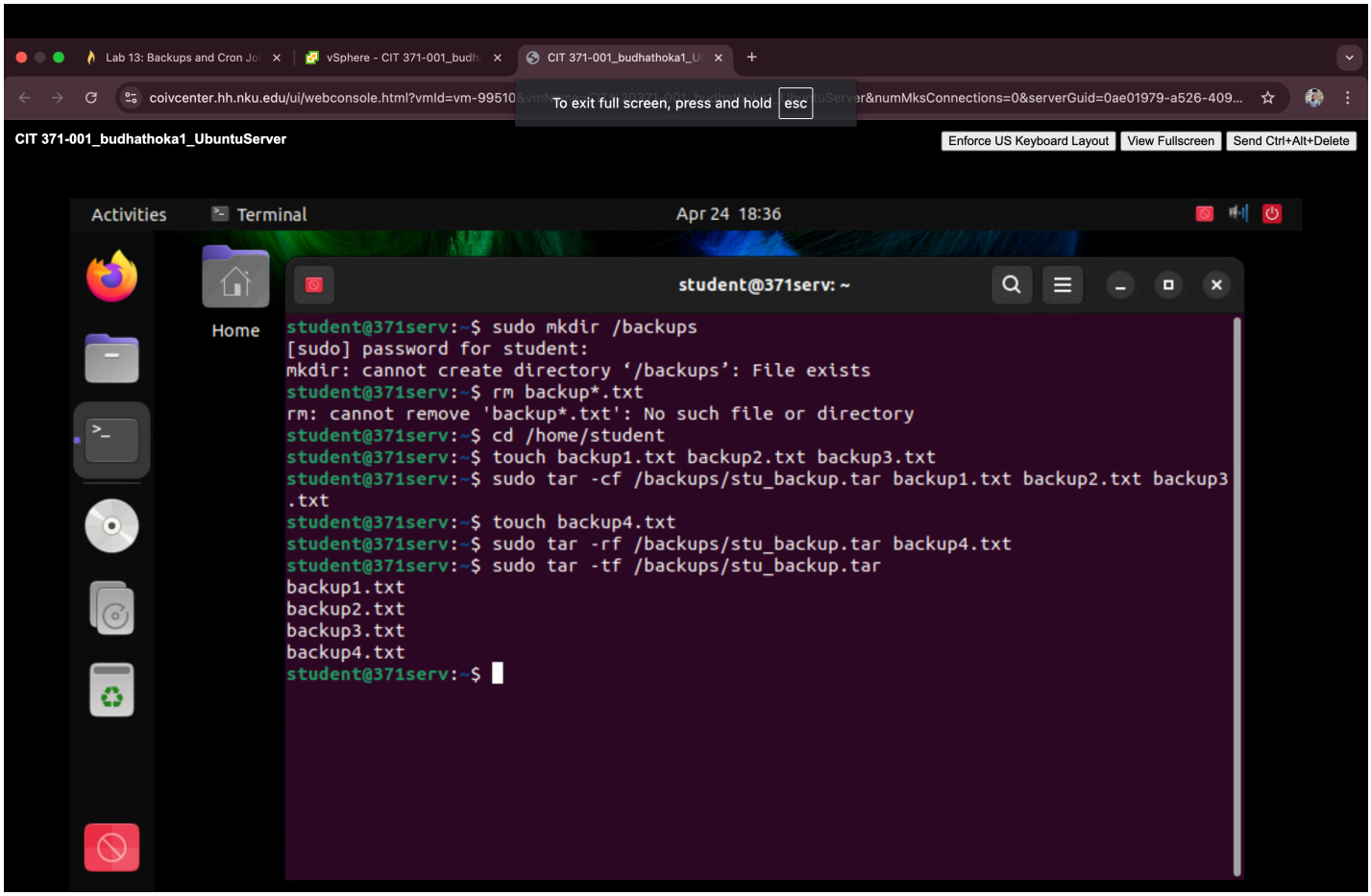
→ tar -rf /backups/stu_backup.tar /home/student/backup4.txt

→ tar -tf /backups/stu_backup.tar

→ tar -cjf /backups/stu_backup.tar.bz2 /home/student/backup*.txt

→ tar -tjf /backups/stu_backup.tar.bz2





Part3: Cron Jobs

→ apt update && apt upgrade -y

→ sudo chmod +x /root/update_system.sh

→ sudo crontab -e

→ 30 2 1 * * /root/update_system.sh

→ echo "---- $(date) ----" >> /root/diskspace.txt

→ df -H >> /root/diskspace.txt

→ sudo chmod +x /root/check_disk.sh

→ 0 8 * * 1 /root/check_disk.sh

→ echo "Time to go home!"

→ chmod +x /home/student/gohome.sh

→ crontab -e

→ 30 16 * * * /home/student/gohome.sh

→ crontab -l

→ sudo crontab -l

```
student@371serv:~$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
# 30 16 * * * /home/student/gohome.sh
```

```
student@371serv:~$ sudo crontab -l
[sudo] password for student:
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
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# Notice that tasks will be started based on the cron's system
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# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
# 30 2 1 * * /root/sysupdate.sh
# 0 8 * * 1 /root/checkspace.sh
```

Conclusion:

This lab provided hands-on experience with two critical Linux system administration tools: tar and crontab. By practicing file backups using tar archives and automating system tasks with cron jobs, I gained a deeper understanding of how Linux systems can be managed efficiently and reliably. These skills are essential for maintaining system integrity, ensuring data is preserved, and reducing the need for manual intervention in routine tasks. Overall, the lab demonstrated how powerful and flexible Linux can be when used with the right tools and proper planning.

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